

Linear Regression Inference

Statistics on a linear model

Download the section 12 .Rmd handout to
STAT240/lecture/sect12-regression-inference.

Download the files
lake-monona-winters-2024.csv, riley.txt
and lions.csv to STAT240/data.

We've seen how to estimate a linear model, but we have not done any statistics.

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

Is there an actual linear relationship between x and y ? There is when the slope β_1 is nonzero.

We will extend our point estimate for slope into an interval estimate for $\hat{\beta}_1$.

$$\hat{\beta}_1 \pm \frac{\alpha}{2} \text{ Critical value} \times \text{Standard error of } \hat{\beta}_1$$

This is a $1 - \alpha$ confidence interval for the slope.

What is the estimation error of $\hat{\beta}_1$?

$$\hat{se}(\hat{\beta}_1) = \frac{S}{\sqrt{(n-1)s_X^2}}$$

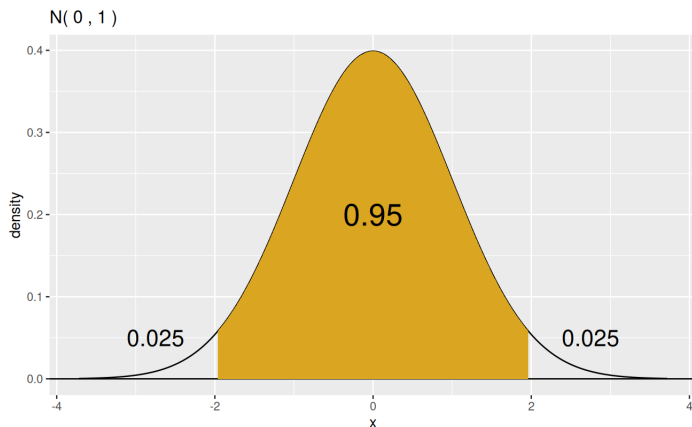
- Numerator: estimate for σ
- Denominator: variability of X

How do we guarantee $1 - \alpha$ coverage? Use a quantile on the sampling distribution.

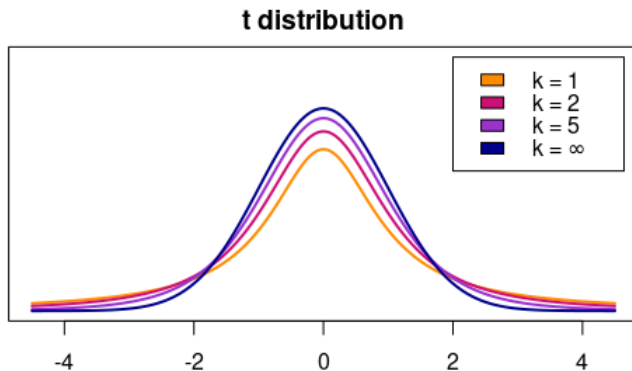
The sampling distribution for $\hat{\beta}_1$ is related to the **Student's T distribution**.

The T is similar to $N(0, 1)$.

For 95% confidence:



What does the T look like?



The T has heavier tails than $N(0, 1)$, controlled by degrees of freedom.

In simple linear regression, $df = n - 2$.

Find critical values (quantiles) with `qt`.

$$\hat{\beta}_1 \pm t_{\alpha/2, n-2} \times \text{Standard error of } \hat{\beta}_1$$

Find these values with the `lm` summary.

A 95% CI for slope in the height model is:

$$(0.244, 0.256)$$

$$\hat{\beta}_1 \pm t_{\alpha/2, n-2} \times \text{Standard error of } \hat{\beta}_1$$

Build and interpret a 98% CI for the slope of the Lake Monona linear model.

- Use qt to find the critical value

Are we confident that year and duration are related?

We can also test the slope.

The formal hypothesis testing procedure is:

- Write **hypotheses** about parameter
- Calculate **test statistic**
- Identify **null distribution**
- Calculate **p-value** on the null

The test statistic is the evidence against the null hypothesis in our data. Usually looks like this:

$$\frac{\text{Estimated value} - \text{Value under null}}{\text{Estimation error}}$$

If the null is true, the test statistic is close to 0.

The p-value is the probability of seeing our data or something more extreme, under the null.

Which direction is “more extreme?”

First, let's test whether the slope of the Lake Monona model is negative, with $\alpha = 0.05$.

$$H_0 : \beta_1 \geq 0 \quad \text{versus} \quad H_A : \beta_1 < 0$$

For a slope test, our test statistic is

$$T = \frac{\hat{\beta}_1 - \beta_{null}}{\hat{se}(\hat{\beta}_1)}$$

If H_0 is true and $\beta_1 \geq 0$, then T follows a T distribution with $n - 2$ degrees of freedom.

We have

$$t_{obs} = \frac{-0.223 - 0}{0.02667} = -8.36$$

Is this value consistent with a t_{n-2} distribution?
No - very small p-value in “less than” direction

What if we just want to know if x and y are related?

$$H_0 : \beta_1 = 0 \quad \text{versus} \quad H_A : \beta_1 \neq 0$$

We would have the same test statistic, but a different p-value.

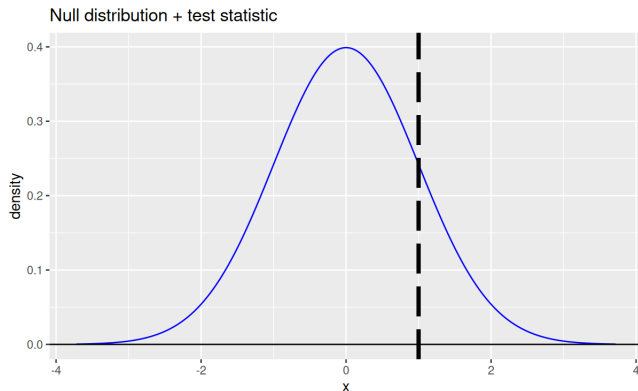
This is also given in the `lm` output.

Do x and y have a linear relationship?

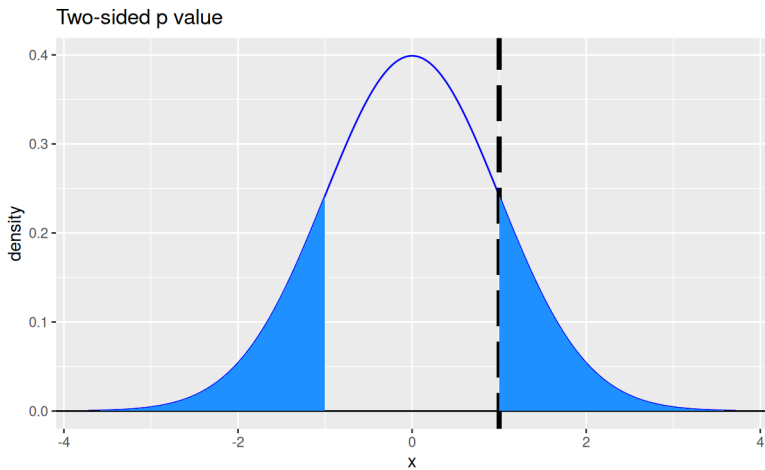
$$H_0 : \beta_1 = 0 \quad \text{versus} \quad H_A : \beta_1 \neq 0$$

This is a **two-sided** test.

- p-value: outcomes more extreme than test statistic on both sides



Suppose we have a positive test statistic.

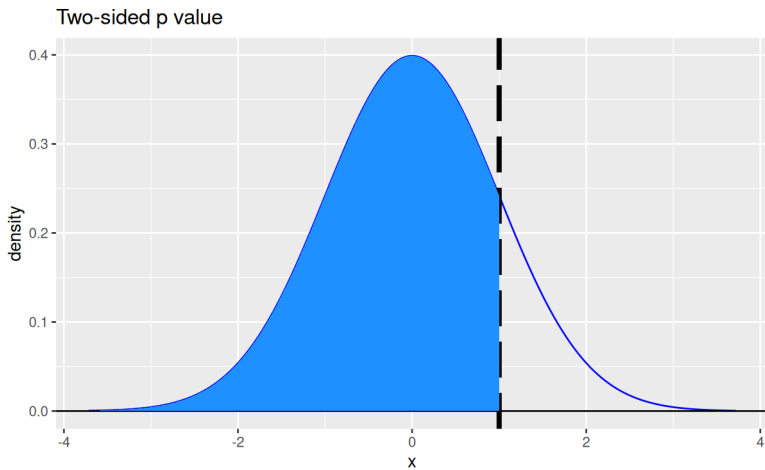


Do x and y have a negative relationship?

$$H_0 : \beta_1 \geq 0 \quad \text{versus} \quad H_A : \beta_1 < 0$$

This is a **one-sided** (negative) test.

- p-value: outcomes less than test statistic

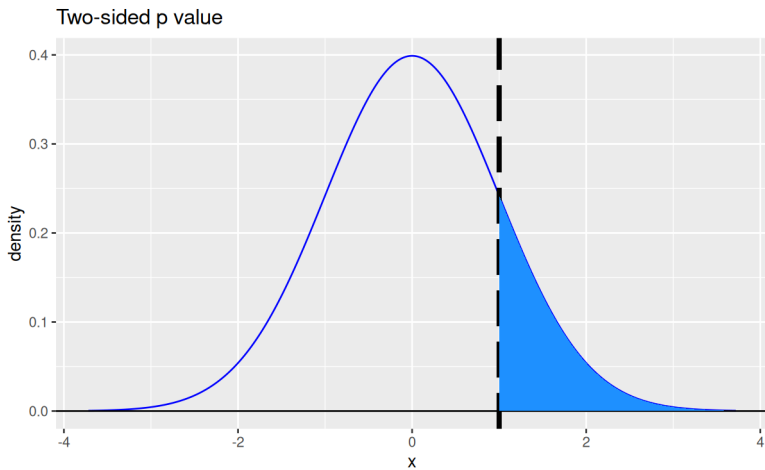


Do x and y have a positive relationship?

$$H_0 : \beta_1 \leq 0 \quad \text{versus} \quad H_A : \beta_1 > 0$$

This is a **one-sided** (positive) test.

- p-value: outcomes greater than test statistic



We've seen how to predict a value with a linear model. Let's turn that into an interval.

Predicted value $\pm$ Critical value $\times$ Prediction error

In the lion ages data, we want to relate a lion's age to the % of its nose that is black.



Six traits can be used to accurately estimate a lion's age: nose darkness, mane growth (in males), facial scarring, teeth color and wear, and jaw slackness. Due to variance between individuals, age should be estimated based on multiple characteristics.

We predict a 5-year-old lion to have a 36% black nose. Formally,

$$\left(\hat{y} \mid x^* = 5 \right) = 0.36$$

$$(\text{Fitted value given } x^*) = \hat{\beta}_0 + \hat{\beta}_1 x^*$$

The uncertainty in this point estimate depends on what exactly we are predicting.

- Predicting the position of the line itself. This is the *average* nose % for all 5-year-old lions.
- Predicting the nose % for a *single* 5 year old lion.

The first type of prediction, the position of the line, is $E(\hat{y} \mid x^*)$.

Let's investigate this with simulation.

- Generate n random points from $\beta_0 + \beta_1 x + \epsilon$
- Calculate $\hat{\beta}_1$ and $\hat{\beta}_0$ and plot the line

The estimated standard error of the position of the line is

$$\hat{se}(E(\hat{y} \mid x^*)) = S \sqrt{\frac{1}{n} + \frac{(x^* - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$$

The critical value for our CI is the same as before. It is a $\alpha/2$ critical value from the T with $n - 2$ degrees of freedom.

$$\hat{y}|x^* \pm t_{\alpha/2, n-2} \times S \sqrt{\frac{1}{n} + \frac{(x^* - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$$

This is what `geom_smooth` is doing!

Use `predict` to calculate the CI for us.

Set `interval = "confidence"`. We can also plot this against the data.

The uncertainty in this point estimate depends on what exactly we are predicting.

- Predicting the position of the line itself. This is the *average* nose % for all 5-year-old lions.
- Predicting the nose % for a *single* 5 year old lion.

The second type of prediction is $\hat{y} \mid x^*$.

Again, the point estimate is just found by plugging x^* into the model.

This type of prediction has more error than predicting the position of the line.

The estimated standard error of a new prediction is

$$\hat{se}(\hat{y} \mid x^*) = S \sqrt{1 + \frac{1}{n} + \frac{(x^* - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$$

We have an extra $+1$ term for applying our model to a new data point.

This gives us a **prediction** interval.

$$\hat{y}|x^* \pm t_{\alpha/2, n-2} \times S \sqrt{1 + \frac{1}{n} + \frac{(x^* - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$$

Again, use predict.

Confidence interval: position of the line
Prediction interval: new y value

- PIs are wider than CIs
- Both intervals are wider when we are further from $\bar{x}$

The **coefficient of determination** R^2 is

$$R^2 = \frac{\text{Total variability of } y - \text{Model error}}{\text{Total variability of } y}$$

R^2 is the fraction of the total variability in y explained by x (via the regression line).

We can find R^2 in the summary output of an `lm` object in R.

If we only have two variables (x and y), then R^2 is equal to the square of the correlation coefficient.

$$R^2 = r^2$$

(This does not necessarily hold for more complex models).

R^2 is a useful measure of how well x explains y , but it does not help us in evaluating assumptions.